

**Conclusion:** Neoadjuvant chemotherapy followed by hypofractionated radiotherapy in locally advanced unresectable OCSCC shows acceptable acute radiation toxicity and good efficacy.

**Abstract 2696 – Table 1: Patient and disease characteristics**

| Patient Characteristics                                                             | Distribution (n=10)     |
|-------------------------------------------------------------------------------------|-------------------------|
| Median Age (Range)                                                                  | 54 years (Range 18-60)  |
| Male: Female                                                                        | 8:2                     |
| Median KPS (Range)                                                                  | 90 (80-90)              |
| Subsite: Buccal mucosa Tongue other                                                 | 4 (40%) 5 (50%) 1 (10%) |
| Differentiation: Well differentiated<br>Moderate differentiated Poor differentiated | 5(50%) 5(50%) 0         |
| Stage Group AJCC 8 <sup>th</sup> III:IV                                             | 1:9(10%v/s90%)          |
| Abbreviations: KPS=Karnofsky performance status                                     |                         |

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## 2697

### NANORAY-312: A Phase III Pivotal Study of NBTXR3 Activated by Investigator's Choice of Radiotherapy Alone or Radiotherapy in Combination with Cetuximab for Platinum-Based Chemotherapy-Ineligible Elderly Patients with Locally Advanced HNSCC

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**Purpose/Objective(s):** Elderly patients with locally advanced head and neck squamous cell carcinoma (LA-HNSCC) are largely underrepresented in clinical trials (CT). This population lacks broad access to CTs due to an increased prevalence of comorbidities and age-related conditions. However, as population aging increases, more elderly patients will be diagnosed with LA-HNSCC although many will not be eligible to receive standard cisplatin-based concurrent chemoradiation (CRT). Toxicities, treatment burden and compliance associated with cisplatin are a concern. NBTXR3, a first-in-class radioenhancer, is composed of functionalized hafnium oxide nanoparticles, administered by a one-time intratumoral injection and activated by radiotherapy (RT). It has EU marketing approval for preoperative treatment of locally advanced soft tissue sarcomas. NBTXR3 locally amplifies tumor-killing effects of radiotherapy without adding toxicity to surrounding healthy tissue. NBTXR3 has been shown to prime the adaptive immune response in cancer models. In a Phase I trial of NBTXR3 + RT alone in elderly LA-HNSCC patients, this combination was shown to be safe and an Objective Response Rate (ORR) of 85.4% and CRR of 51.2% was reported at ASTRO 2021. A pivotal randomized phase III study is now recruiting.

**Materials/Methods:** NANORAY-312 is a global, open-label, randomized (1:1), 2-arm, Phase III (Pivotal Stage) study to investigate the efficacy and safety of NBTXR3/RT ± cetuximab versus RT ± cetuximab (Investigator's choice) in treatment-naïve, platinum-ineligible, elderly patients with LA-HNSCC. 500 patients will be enrolled. Eligible patients are ≥65 years with at least one measurable (RECIST 1.1) biopsy-confirmed T3-4 (AJCC v8) SCC of the oral cavity, oropharynx, hypopharynx, or supraglottic larynx

which is amenable for intratumoral injection. Patients will receive IMRT (70 Gy in 35 fractions / 7 weeks). For patients with oropharyngeal cancer, human papilloma virus (HPV) status must be declared. In addition to the primary tumor, a single accessible LN in the neck is considered an eligible site for intranodal injection.

**Results:** The primary endpoint is PFS. The key secondary endpoint is OS. Other secondary endpoints include ORR, safety and tolerability, and quality of life. Exploratory endpoints include tumor immune response biomarkers, and radiographic change in the size of the tumor after 50 Gy of RT. Enrollment began in January 2022 and continues.

**Conclusion:** NBTXR3/RT ± cetuximab is under investigation in elderly patients with LA-HNSCC, a population which is underrepresented in clinical trials and with limited treatment options. This study has the ClinicalTrials.gov identifier NCT04892173.

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## 2698

### Proton Therapy for Uveal Melanoma: A Single-Institution Experience

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**Purpose/Objective(s):** Uveal melanoma (UM) is the most common intraocular malignancy in adults. Eye-preserving approaches like episcleral plaque brachytherapy are common but limited in patients (pts) with large tumors or those adjacent to the optic nerve. Proton therapy (PT) offers an alternative eye-preserving treatment option. We report a single-institution experience utilizing PT to treat UM. We hypothesized that PT would provide excellent local control (LC) and eye preservation (EP) with acceptable toxicity and visual outcomes across all patients, including those with large tumors in adverse locations.

**Materials/Methods:** We reviewed the records of treatment-naïve pts who received PT for non-metastatic UM between June 2012 and December 2020. All pts had ≥ 6 months of follow-up. Treatment planning was performed in EyePlan (Clatterbridge, UK) based on eye/tumor measurements and 3-4 implanted scleral fiducial markers, registered fundus photographs, and confirmed by CT/MRI. Pts were set-up in a seated position with gaze fixation and real-time image guidance using a single en face beam to a total 50-60 Gy RBE in 4-5 consecutive daily fractions. We estimated disease control and survival, including LC, EP, distant metastasis-free survival (DMFS), cause-specific survival (CSS), overall survival (OS), and grade 3+ toxicity as defined by CTCAEv5 using the cumulative incidence method. Regression of multiple variables simultaneously was accomplished with competing-risks Cox regression.

**Results:** 120 pts met inclusion criteria; 33 had a biopsy of their lesion; 9.2% were class 2. Moreover, 47.5% had COMS medium while 35.8% had large tumors. Median tumor thickness was 5.0 mm (0.5 – 13.0) and median basal dimension was 13.0 mm (4.0 – 25.0). Median distance to the fovea and optic disc was 3.5 mm. 38% had retinal detachment with 47% having significantly impaired visual acuity at baseline (worse than Snellen 20/40). 98.3% received 60 Gy RBE over 4 daily fractions. Median follow-up was 31 months (6 – 104). The 2-year estimates (95% CI) for LC, EP, DMFS, CSS, and OS were 94% (88-97%), 95% (90-98%), 88% (81-93%), 94% (88-97%), and 90% (83-95%), respectively. 9 pts underwent enucleation following PT